

Gent Vigilon

Diagnostics & System Interrogation

Viewing Event Logs (Active & Historic)& Exception Codes

Active Events: [Info] -> [Display] -> [Active] -> Select category -> [Enter]

Historic Fires: [Info] -> [Events] -> [HistFire] -> "1 THRU 255" -> [Enter]

Checking Sensor Exception/Condition Codes: [Info] -> -> [Status] -> [Device] -> Enter Address -> [Loop] -> Loop No -> [Enter]

Condition codes range 0-3. Codes 4 & 5 represent chamber drift. A value of 3 means 'Out of Limits' [Clean/Replace].

Checking PSU Readings : [Test/Eng] -> [Test] -> [PSU]

Interrogating Info & Status

View Loop Map (Cabling order): [Info] -> [Display] -> -> [Loop Map] -> Enter Loop No-> [Enter]

Interrogate Device Status: [Info] -> -> [Status] -> [Device] -> Enter Address -> [Loop] -> Loop No -> [Enter]

View Device Labels: [Info] -> [Display] -> [Label] -> [Device] -> Enter Address -> [Loop] -> Loop No -> [Enter]

Card Status & Roles:

- Card 0 (MCC): Main Control Card. Manages panel times, configurations, serial links, and networks.
- Cards 1-6 (LPC): Loop Processor Cards. Controls communication and addressing on active loop circuits.
- Card 13 (Terminal): Base terminal board handling clean contacts, clean inputs, and conventional sounder lines.
- Card 14 (NVM): Non-Volatile Memory. Stores configuration checksum backups.
- Card 15 (Network): Network Card linking panel nodes on copper or fiber networks.

Checking the Status of Sectors, Builds, Devices and Sector Links

Check Sector Links: [Info] -> [Status] -> [SectLink] -> [Enter Sector] -> [Loop] -> [Enter Loop] -> [Enter]

Check Command Build Status: [Info] -> -> [Status] -> [Build] -> Enter Build No -> [Enter]

Check Sector Status: [Info] -> -> [Status] -> [Sector] -> Enter Sector No -> [Enter]

Interrogate device: [Info] -> -> [Status] -> [Device] -> Enter Address -> [Loop] -> Loop No -> [Enter]

How to Interrogate and Interpret the Device Status Screen

LB1:C / LB2:C: Loop Breakers 1 and 2 are Closed (healthy) or Open (isolator tripped)

End1:Hi / End2:Hi: High-frequency communication levels are healthy. If Lo or Marginal, check for high resistance.

I/O: ...0: Displays active input/output channels on interfaces (e.g. ...0 represents channel 4 as output, I... represents channel 1 as input).

Anal: Displays active analog sensor chambers (e.g. 1, 2 is Dual Optical, 4 is heat).

Finding Devices

Turn LED ON: [Test/Eng] -> PIN -> -> [Outputs] -> [Device] -> Enter Address -> [Loop]-> Loop No -> [Line] -> "4" -> Select [ON] -> [Enter]

Turn LED OFF: [Test/Eng] -> PIN -> -> [Outputs] -> [Device] -> Enter Address -> [Loop]-> Loop No -> [Line] -> "4" -> Select [OFF] -> [Enter]

Physically Finding Devices: [Test/Eng] -> PIN -> [Loop] -> [Find Dev] -> [Start] -> Enter Device Address -> [Enter] (*Pulsates the Local LED/sounder; use [Next]/[Previous] keys to scan*)

Isolations & Fault Clearing

Disablements & Safe Isolations:

Sectors: [Control] -> [Disable] -> [Sector] -> "1 THRU 32" -> [Loop] -> "1" -> [Enter] (*Enable with [Enable]*)

Aux Relays: [Control] -> [Disable] -> [Aux Rly] -> "1 THRU 2" -> [Enter] (*Enable with [Enable]*)

I/O Lines: [Control] -> [Disable] -> [IO Line] -> "1" -> [Device] -> "3" -> [Loop] -> "1" -> [Enter] (*Enable with [Enable]*)

Builds: [Control] -> [Disable] -> -> [Build] -> "1 THRU 255" -> [Enter] (*Enable with [Enable]*)

Master Alarms: [Control] -> [Disable] -> [Malarms] -> [Enter] (*Enable with [Enable]*)

Loop Comms: [Control] -> [Disable] -> -> [Comms] -> "1" -> [Enter] (*Enable with [Enable]*)

Sensors: [Control] -> [Disable] -> [Sensors] -> "1 THRU 207" -> [Loop] -> "1" -> [Enter] (*Enable with [Enable]*)

MCPs: [Control] -> [Disable] -> [MCPs] -> "1 THRU 207" -> [Loop] -> "1" -> [Enter] (*Enable with [Enable]*)
- VERY RARE YOU NEED TO DO THIS

Clear Latching Faults (Allocation/Slave lost): Must run loop re-allocation: [Test/Eng] -> PIN -> [Loop] -> [Allocate] -> Loop No -> [Enter]

Clear Clearable Faults (FLASH corrupt/Checksum/NVM mismatch): [Test/Eng] -> [Config] -> [Clear] -> [Fault] -> [Enter] -> Scroll to target -> Select [Clear]

Hardware Interventions & Loop Management

SAFE addressing & Soft (un-SAFE) addressing:

Un-SAFE (Soft Address): [Set Up] -> PIN -> [SetUp] -> [Device] -> Enter Address -> [Loop] -> Loop No -> [Safe] -> "0" -> [Enter]

Re-SAFE (Lock Address): [Set Up] -> PIN -> [SetUp] -> [Device] -> Enter Soft Address -> [Loop] -> Loop No -> [Safe] -> Enter Target Address -> [Enter]

Stopping & Re-allocating a Loop Card:

Stop Loop: [Test/Eng] -> PIN -> [Loop] -> [Stop] -> "1" -> [Enter]

Re-Allocate Loop: [Test/Eng] -> PIN -> [Loop] -> [Allocate] -> "1" -> [Enter]

Stopping a Loop on older Vigilon Panels: [Test/Eng] -> [Card] -> [Reset] -> "1" -> [Enter] twice in quick succession

Adding and Removing Devices from Active Loops

Un-SAFE old device: [Set Up] -> PIN -> [SetUp] -> [Device] -> Enter Address -> [Safe] -> "0" -> [Enter]

Stop loop: [Test/Eng] -> enter PIN -> [Loop] -> [Stop] -> Enter Loop No -> [Enter]

(Now physically modify the wiring)

Start & allocate: [Test/Eng] -> [Loop] -> [Allocate] -> Enter Loop No -> [Enter] *(Auto-assigns lowest vacant soft address)*

Lock SAFE address: [Set Up] -> PIN -> [SetUp] -> [Device] -> Enter Soft Address -> [Safe]-> Enter Target Address -> [Enter]

Like-for-Like Sensor Replacement

Process for a SAFE Device: Un-safe original address (set to 0) -> stop loop -> swap physical head -> allocate loop -> re-safe original address back onto new head.

Process for a Non-SAFE Device: Stop loop -> swap physical head -> allocate loop (device automatically takes the soft address, no safe mapping required).

Loop Repair: [Menu On/Off] -> [Test/Eng] -> [Loop] -> -> -> [Repair] -> [Loop] -> Enter Loop No -> [Enter]

Disable and Enable Panel Buzzer: [Test/Eng] -> [Config] -> [Buzzer] -> Enter Buzzer type -> [Off/On] -> [Enter]

Setting the System Clock: [Set Up] -> [SetClock] -> Enter Date and Time (Format: HH:MM DD/MM/YY) -> [Enter]

Configurations

Modify Device Label (32-Char Limit): [Menu On/Off] -> [Set Up] -> [Usercode] -> [Set Up] -> -> [Modify] -> [Label] -> [Device] -> Enter Device Label -> [Loop] -> Enter Loop -> [Enter] Type Custom Location Label on Keypad -> [Enter] - BACK UP TO NVM UPON COMPLETION

Enable Long Labels (64-Char Limit, Tokenized): [Menu On/Off] -> [Test/Eng] -> [Usercode] -> -> [Config] -> Press -> [Label] -> Select [On] -> [Enter]

Backing Up and Restoring from the NVM

Software Unprotect the NVM: [Set Up] -> PIN -> -> [Protect] -> [NVM Card] -> [Disable] -> [Enter]

Back up ALL cards (0 to 15) to NVM: [Set Up] -> PIN -> [SetUp] -> [Backup/Restore] -> "0 THRU 15" -> [NVM Card] -> [Enter] * Wait for the Checksum 14 message before proceeding! *

Software Protect the NVM: [Set Up] -> PIN -> -> [Protect] -> [NVM Card] -> [Enable]-> [Enter]

Backing Up and Restoring from the SD Card

Back up ALL site configurations to SD Card: [SetUp] -> [Backup] -> "0 THRU 15" -> [SD Card]-> [Enter]

Restore site configurations from SD Card: [SetUp] -> [Restore] -> "0 THRU 15" -> [SD Card]-> [Enter]

Advanced Programming & Cause/Effect

Zones and Coincidence Detection

Assigning Devices to Zones : [Set Up] -> [Assign] -> [Device] -> Enter Device Number -> [Loop] -> Enter Loop Number -> [Zone] -> Enter Target Zone -> [Enter]

Zone Display Modes:

ZONE (ZONAL): Displays only the general Zone Label (factory default).

1stDev (FIRST OS): Displays only the specific custom label of the very first device that alarmed.

AllDev (ALL OSs): Displays and cycles the custom labels of all active devices in fire within the zone.

Determine Zone mode : [Set Up] -> [SetUp] -> [Zone] -> Select Zone -> [Mode] -> [AllDev/All OS] -> [Enter]

Set up Zone Task : [Set Up] -> [SetUp] -> [Remove] -> [Zone] -> Select Zone -> [Task] -> How many fires/triggers from each Sensor -> [Build] -> Select a Build (this task will be the Trigger for the selected Command Build) -> [Sensors] -> How many differing Sensors to be activated for this to operate -> [Pre-Fires] -> How many pre fires required -> [Fires] -> How many fires required-> [Superfires] -> How many Super Fires required -> [Enter]

Set up the Command Build keeping in mind the Trigger will be the created Zone Task e.g : [Set Up] -> [Set Up] -> [Build] -> Select Build selected in Zone Task -> [Action] -> [Sector] -> "1 - 32" -> [Loop] -> Select Loop -> [Action] -> [Sig3] -> [Reversbl] -> [Enter]

Place Zone in Walk Test : [Control] -> [Test]-> [Zone] -> Enter Zone -> [Loop] -> Enter Loop -> [With/Without Sound] -> [Enter]

Remove Zone from Test Mode : [Control] -> [Active] -> [Zone] -> Enter Zone -> [Loop] -> Enter Loop -> [Enter]

Sectors

By default, Gent panels initialize with a 'one-out, all-out' default fire plan, assigning all devices to Loop Sector 1 to sound a global evacuation (Signal 2) on any alarm. Before building a custom zoning plan, you must strip out these default settings.

CRITICAL FAIL-SAFE RULE: Always retain Sector 1 as a backup. Leave Sector 1 unassigned to any device, but inter-actioned on all other sectors. If any device loses its custom sector mapping during a power cycle or NVM recovery conflict, it automatically defaults back to Sector 1. Leaving Sector 1 actioned guarantees these 'orphaned' devices will still sound under fire conditions.

De-Actioning Sectors

Remove Sensing Devices from Sector 1: [Set Up] -> [Set Up] -> [Remove] -> [Device] -> "1 THRU 207" -> [Loop] -> "1" -> [Sector] -> "1 THRU 32" -> [Enter]

Remove Call Points from Sector 1: [Set Up] -> [Remove] -> [MCP] -> "1 THRU 207" -> [Loop] -> "1" -> [Sector] -> "1 THRU 32" -> [Enter]

Remove Interface I/O Lines from Sector 1: [Set Up] -> [Remove] -> [IO Line] -> "1 THRU 4" -> [Device] -> "1 THRU 207" -> [Loop] -> "1" -> [Sector] -> "1 THRU 32" -> [Enter]

De-Action Sectors (Keep Sector 1 as Backup): [Set Up] -> -> [Deaction] -> [Sector] -> "2 THRU 32" -> [Loop] -> "1" -> [Sector] -> "2 THRU 32" -> [Loop] -> "1" -> [Enter]

Fail-Safe Sector 1 Actions (Cross-Link Sector 1): [Set Up] -> [Action] -> [Sector] -> "1" -> [Loop] -> "1" -> [Sector] -> "2 THRU 32" -> [Loop] -> "1" -> [Signal 2] -> [Enter] [Set Up] -> [Action] -> [Sector] -> "2 THRU 32" -> [Loop] -> "1" -> [Sector] -> "1" -> [Loop] -> "1" -> [Signal 2] -> [Enter]

Assigning Sectors

Assign Device to a Sector : [Set Up] -> [Set Up] -> [Assign] -> [Device] -> Enter Target Device Number -> [Loop] -> Enter Loop Number -> [Sector] -> Enter Target Sector -> [Enter]

Action Sector to Sector for Cause & Effect : [Set Up] -> [Action] -> [Sector] -> ACTION/TARGET SECTOR -> [Loop] -> "1" -> [Sector] -> TRIGGER SECTOR -> [Loop] -> Enter Loop -> Select [Signal 1/2/3] -> [Enter]

By default, pressing 'Silence Alarms' on the panel fascia will restore all active output lines. However, heavy plant systems must never restore automatically when sounders are silenced. They must remain isolated until the panel is formally reset. You can modify the plant sector to override the 'Silence Alarms' key to 'No Action'. This forces the auxiliary outputs to remain tripped until a physical panel Reset is executed. 29 - 32 are already configured by default in this way.

Modify Sector to Ignore Silence Alarms : [Set Up] -> [Sector] -> Enter Sector -> [Loop] -> Enter Loop No. -> -> [Silence] -> [NoAction] -> [Enter]

Phased Evac Via Delay Blocks

Example Strategy: A fire on the Top Floor (Sector 12) immediately sounds an evacuation tone on that floor. It initiates Delay Block 1 (8-second delay) to escalate the Middle Floor (Sector 11) from an Alert (Signal 1) to an Evacuate (Signal 2) tone. Simultaneously, Delay Block 1 triggers Delay Block 2 (another 8-second delay) which subsequently escalates the Ground Floor (Sector 10) to full evacuation. First, we need to create the Cause & Effect to ensure that if a fire is detected on a floor, any floor below is sent a Signal 1 (Alert) and the floor in fire and any above receive a Signal 2 (Evac).

Process:

Create Cause and Effect basis for differing floors

1. Assign Sectors

GF: [Set Up] -> [Assign] -> [Device] -> Enter Devices for GF -> [Loop] -> Enter Loop -> [Sector] -> Enter GF Sector ("10") -> [Enter]

1F: [Set Up] -> [Assign] -> [Device] -> Enter Devices for 1F -> [Loop] -> Enter Loop -> [Sector] -> Enter 1F Sector ("11") -> [Enter]

2F: [Set Up] -> [Assign] -> [Device] -> Enter Devices for 2F -> [Loop] -> Enter Loop -> [Sector] -> Enter 2F Sector ("12") -> [Enter]

2. Determine Cause & Effect Plan by Actioning the Sectors (work from top floor down)

This will ensure that if a fire is detected on a floor, any floor below is sent a Signal 1 (Alert) and the floor in fire and any above receive a Signal 2 (Evac).

2F Evac Signal : [Set Up] -> [Action] -> [Sector] -> (ACTION SECTOR) "12" -> [Loop] -> "1" -> [Sector] -> (TRIGGER SECTOR) "12" -> [Loop]-> "1" -> [Signal 2] -> [Enter]

2F Alert Signal to lower floors : [Set Up] -> [Action] -> [Sector] -> (ACTION SECTOR) "10 - 11" -> [Loop] -> "1" -> [Sector] -> (TRIGGER SECTOR) "12" -> [Loop]-> "1" -> [Signal 1] -> [Enter]

1F Evac Signal : [Set Up] -> [Action] -> [Sector] -> (ACTION SECTOR) "11 - 12" -> [Loop] -> "1" -> [Sector] -> (TRIGGER SECTOR) "11" -> [Loop]-> "1" -> [Signal 2] -> [Enter]

1F Alert Signal to lower floors : [Set Up] -> [Action] -> [Sector] -> (ACTION SECTOR) "10" -> [Loop] -> "1" -> [Sector] -> (TRIGGER SECTOR) "11" -> [Loop]-> "1" -> [Signal 1] -> [Enter]

GF Evac Signal to all floors : [Set Up] -> [Action] -> [Sector] -> (ACTION SECTOR) "10 - 12" -> [Loop] -> "1" -> [Sector] -> (TRIGGER SECTOR) "10" -> [Loop]-> "1" -> [Signal 2] -> [Enter]

3. Create Phased Evac using these Sectors and Delay Blocks

This will now cascade an Alarm signal to an Evac signal after an 8 second delay to Lower floors. We only need to create two delay blocks as we have already programmed the action plan in the sectors as above.

Configure Delay Block 1 which will escalate 1F(Sector 11) to Evac

Configure Delay to 8 seconds : [Set Up] -> [SetUp] -> [Delay Block] -> Select Delay Block "1" -> [Delay] -> "00-08" -> [Enter]
Set Signal Action for Delay Block : [Set Up] -> [SetUp] -> [Delay Block] -> "1" -> [Action] -> Select [Sig.2] -> [Enter]

Trigger Delay Block from 2f Sector : [Set Up] -> [Action] -> [Delay Block] -> "1" -> [Sector] -> 2F Sector "12" -> [Loop] -> "1" -> [Enter]

Trigger 1F Sector from Delay Block 1: [Set Up] -> [Action] -> [Sector] -> 1F Sector "11" -> [Loop] -> "1" -> [Delay Block] -> "1" -> [Enter]

Configure Delay Block 2 which will escalate GF(Sector 10) to Evac

This will cascade to the GF from a fire on 2F (the timer for the GF Evac Signal will only start once 1F has gone into Evac)

Configure Delay to 8 seconds : [Set Up] -> [SetUp] -> [Delay Block] -> Select Delay Block "2" -> [Delay] -> "00-08" -> [Enter]

Set Signal Action for Delay Block : [Set Up] -> [SetUp] -> [Delay Block] -> "2" -> [Action] -> Select [Sig.2] -> [Enter]

Trigger Delay Block 2 from Delay Block 1 : [Set Up] -> [Action] -> [Delay Block] -> "2" -> [Delay Block] -> "1" -> [Enter]

Trigger GF Sector from Delay Block 2 : [Set Up] -> [Action] -> [Sector] -> GF Sector "10" -> [Loop] -> "1" -> [Delay Block] -> "2" -> [Enter]

To completely clear all phased evacuation programming from the system

Zero all Delay Timers : [Set Up] -> [Set Up] -> [Delay Block] -> "1 - 16" -> [Delay] -> "00:00" -> [Enter]

De-Action Delay Blocks from Sectors : [Set Up] -> [Deaction] -> [Sector] -> "1 - 32" -> [Loop] -> "1" -> [Delay Block] -> "1 - 16" -> [Enter]

De-Action Sectors from Delay Blocks : [Set Up] -> [Deaction] -> [Delay Block] -> "1 - 16" -> [Sector] -> "1 - 32" -> [Loop] -> "1" -> [Enter]

Sensor States

Setting Sensor Sensitivity State Permanently: [Set Up] -> PIN -> [SetUp] -> [Device] -> Enter Address -> [Loop] -> Enter Loop No -> -> [State] -> "12" -> "12" -> [Enter] (*Sets both Day and Night to State 12 Heat-Only*)

Timed Sensitivity States

To reduce daytime false alarms while maintaining maximum night time protection, you can program individual sensors to automatically switch sensitivity states. This utilizes 16 available Time Slots linked to 15 Time Blocks.

Scenario: *Configure a sensor to operate in State 12 (Heat-Only, making it completely immune to daytime steam/vapes) from 08:30 to 17:30, Monday through Friday, automatically dropping to State 0 (Standard High-Sensitivity Optical-Heat) at all other times and on weekends.*

Program Time Slot 1 (Office hours 08:30 to 17:30, Mon-Fri): [Set Up] -> PIN -> -> [TimeSlot] -> "1" -> [Enable] -> "08:30" -> [Disable] -> "17:30" -> Select [And] -> Select days: [Mon] [Tue] [Wed] [Thu] [Fri] -> [Enter]

Link Time Slot 1 to Time Block 1: [Set Up] -> PIN -> -> [TimeBlock] -> "1" -> [Assign] -> [TimeSlot] -> "1" -> [Enter]

Apply Dual Sensitivity States to the Sensor and Bind to the Time Block : [Set Up] -> PIN -> -> [Device] -> Enter Device -> [Loop] -> "1" -> -> [State] -> "12" -> "0" -> Select [TimeBlock] -> "1" -> [Enter]

Command Builds & Custom Interface Control

Configuring a Single Channel Key Switch to isolate an Ansul system using Command Builds.

In this example, we have installed the isolator (Device 14) and an interface for an input from an Ansul system (Device 27) onto an existing loop. We need to be able to isolate the Ansul for future maintenance via the isolation key switch.

Action a command build with the trigger being the key switch being turned : [Set Up] -> [Set Up] -> [Build] -> Select Build -> [Trigger] -> [I/O Line] -> "1" -> [Device] -> "14" -> [Loop] -> "1" -> [Enter]

Action the Disablement of the Ansul Interface using the Command Build : [Set Up] -> [Set Up] -> [Build] -> Select Build -> [Action] -> [Disable] -> [Device] -> "27" -> [Loop] -> "1" -> [Reversbl] -> [Enter]

Label the Build : [Set Up] -> [Set Up] -> [Build] -> Select Build -> [Label] -> "Ansul Isolated" -> [Enter]

Configurable door buttons U1 to U4 Default Logic

U1 Key (Command Build 251): We will map all sectors 1 THRU 32 as the example below

U2 Key (Command Build 252): We will map Command Builds 2, 3, and 4 simultaneously to isolate loop sectors, conventional master alarms, and auxiliary relays in the example below. In this example, Command Builds 2 - 4 have already been programmed.

U3 & U4 Keys (Command Builds 253 & 254): Predefined for user-customized actions with door LEDs.

Program U1 (CB 251) to Disable All Loop Sectors (Reversible) : [Set Up] -> PIN -> [SetUp] -> [Build] -> "251" -> [Action] -> [Disable] -> [Sector] -> "1 THRU 32" -> [Loop] -> "1" -> [Reversbl] -> [Enter]

Program U2 (CB 252) to Link Support Builds 2, 3, and 4: [Set Up] -> PIN -> [SetUp] -> [Build] -> "252" -> [Action] -> [Build] -> "2 THRU 4" -> Select [ON] -> [Reversbl] -> [Enter]

Class Change

Configure 5-Second Pulsed Lesson Change Class-Bells (09:00, 11:00, 12:30, 14:00, 15:30)

Program Timeslot 1 (09:00:00 to 09:00:05): [Set Up] -> [TimeSlot] -> "1" -> [Enable] -> "09:00:00" -> [Disable] -> "09:00:05" -> [And] -> [Mon] [Tue] [Wed] [Thu] [Fri] -> [Enter]

(Repeat for Slots 2-5 for other times)

Link Slots 1-5 to Time Block 2: [Set Up] -> PIN -> -> [TimeBlok] -> "2" -> [Assign]-> [TimeSlot] -> "1 - 5" -> [Enter]

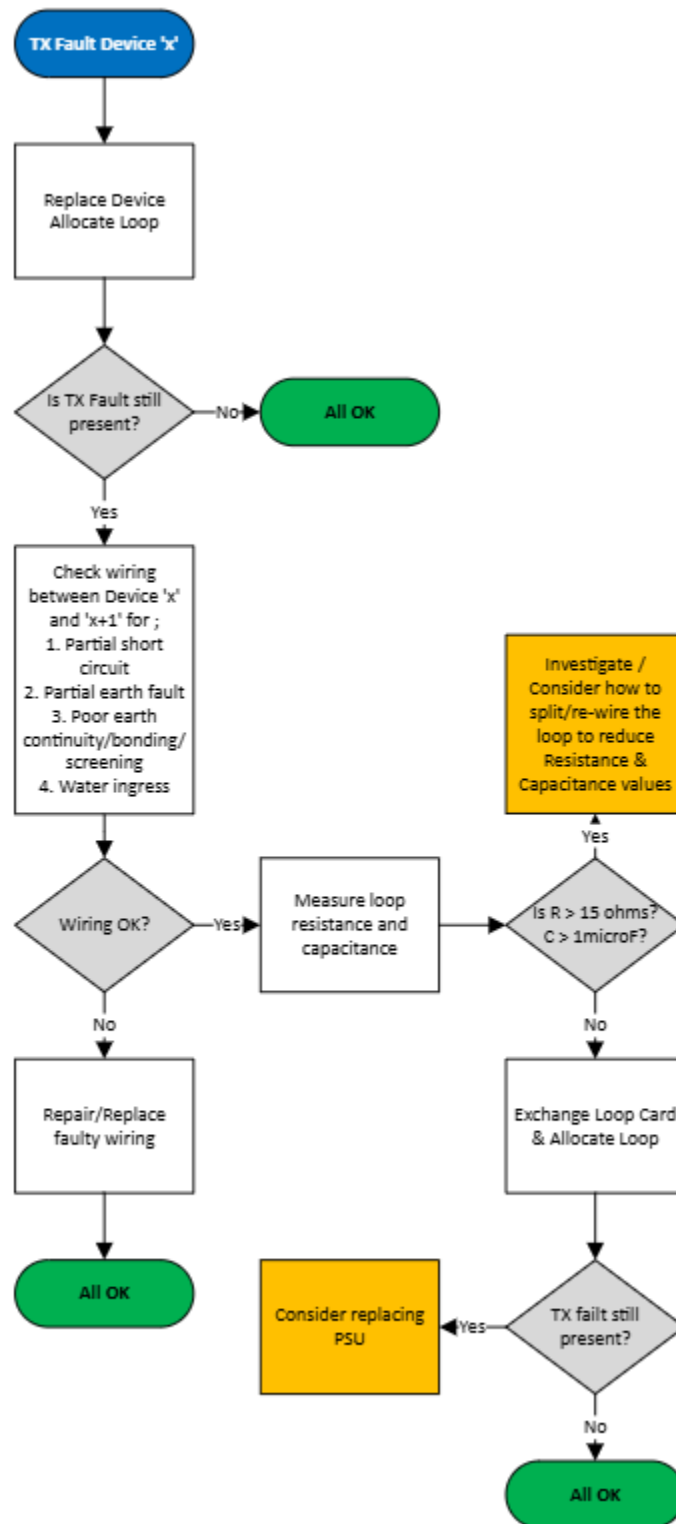
Link Time Block 2 to Trigger Command Build 5: [Set Up] -> [SetUp] -> [Build] -> "5" -> [Trigger] -> [TimeBlk] -> "2" -> [Enter]

Assign Reversible Action (Signal 1 Alert across Sectors 2-5): [SetUp] -> [Build] -> "5" -> [Action] -> [Sector] -> "2 THRU 5" -> [Loop] -> "1" -> [Signal 1] -> [Reversbl] -> [Enter]

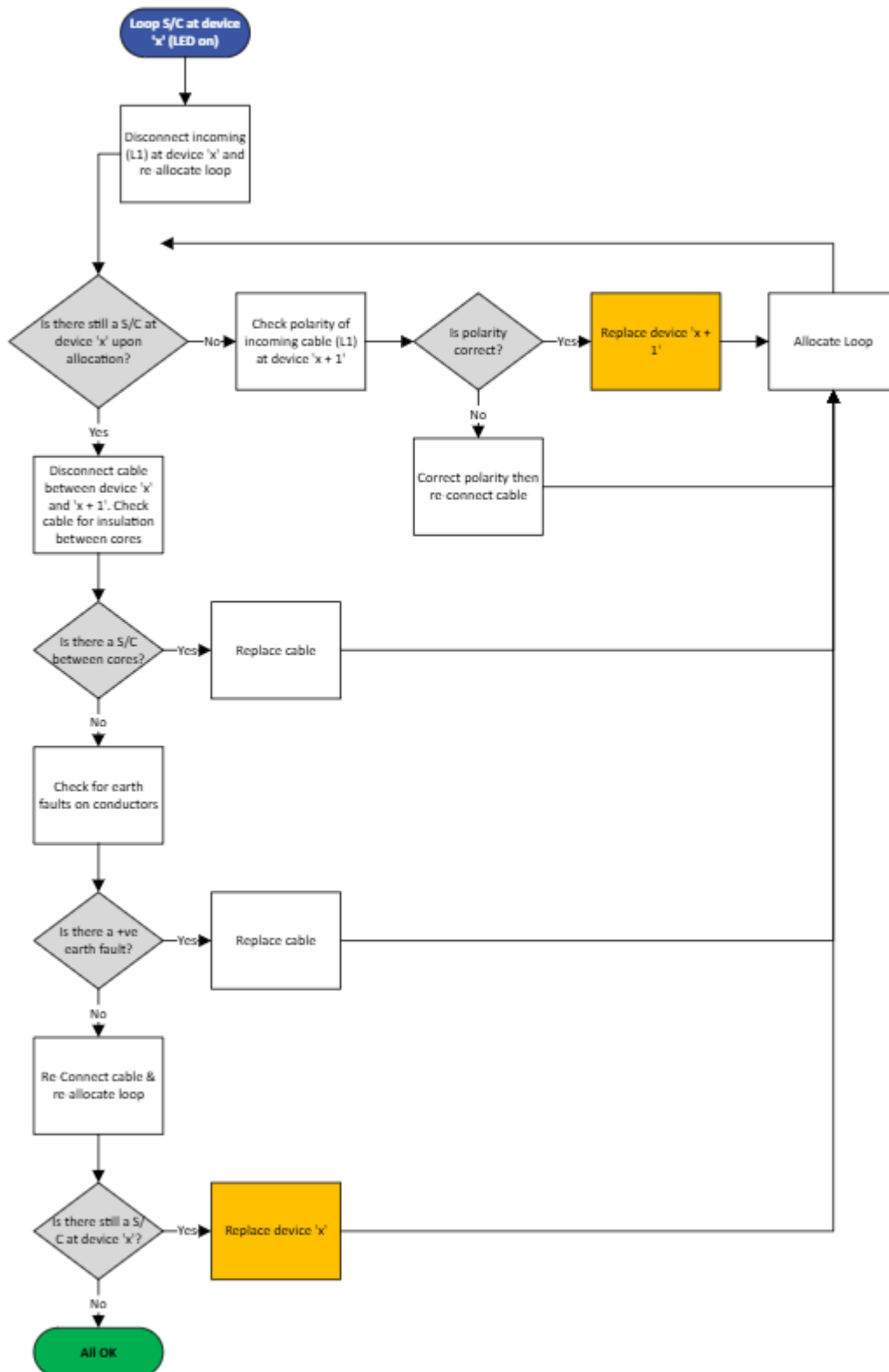
Assign Device to Groups: [Set Up] -> PIN -> -> Select [Device] -> Enter Address -> [Loop] -> Enter Loop No -> Select [Group] -> Enter Group -> [Enter]

Fault Finding Flow Charts

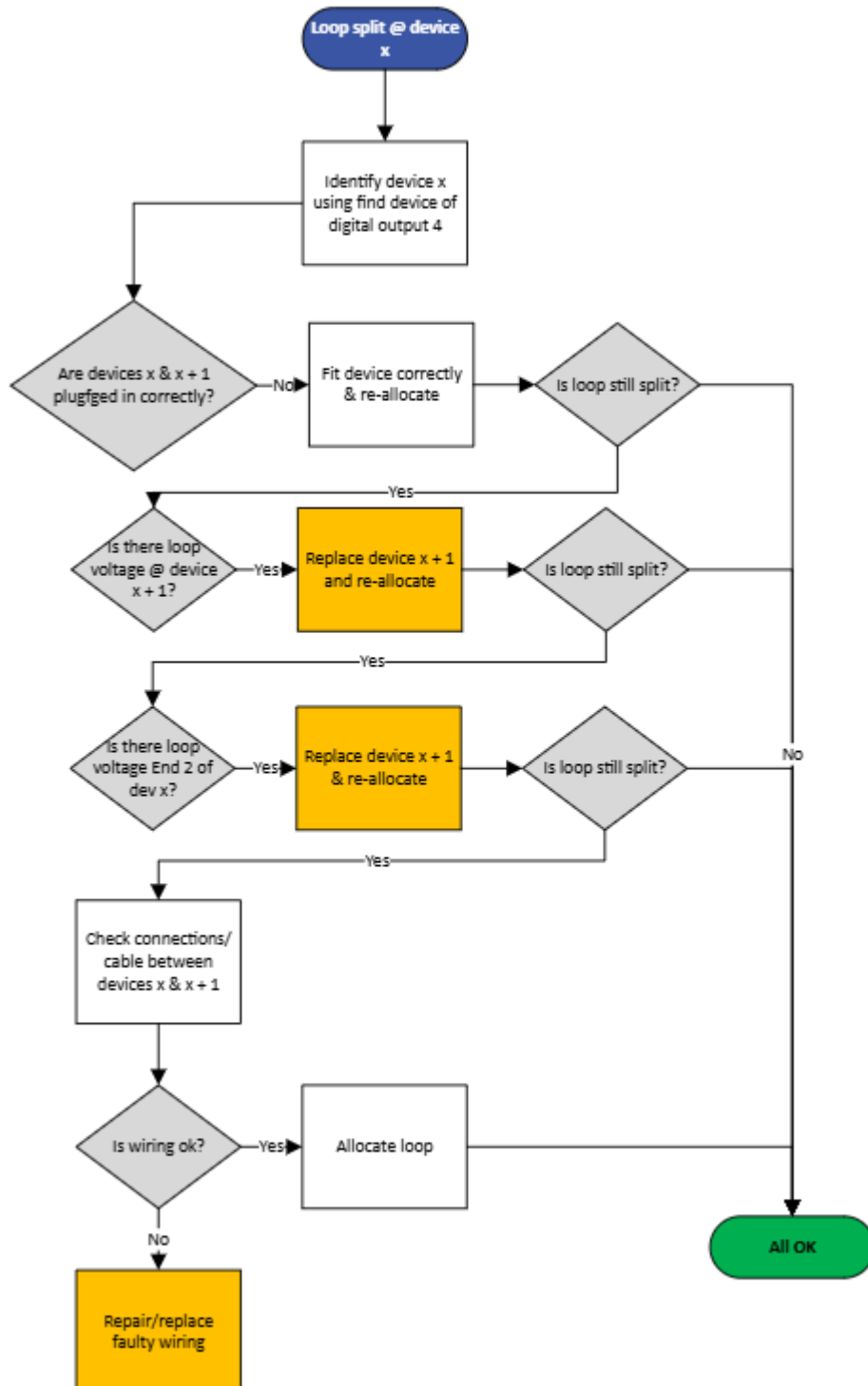
Transmission (TX) Fault



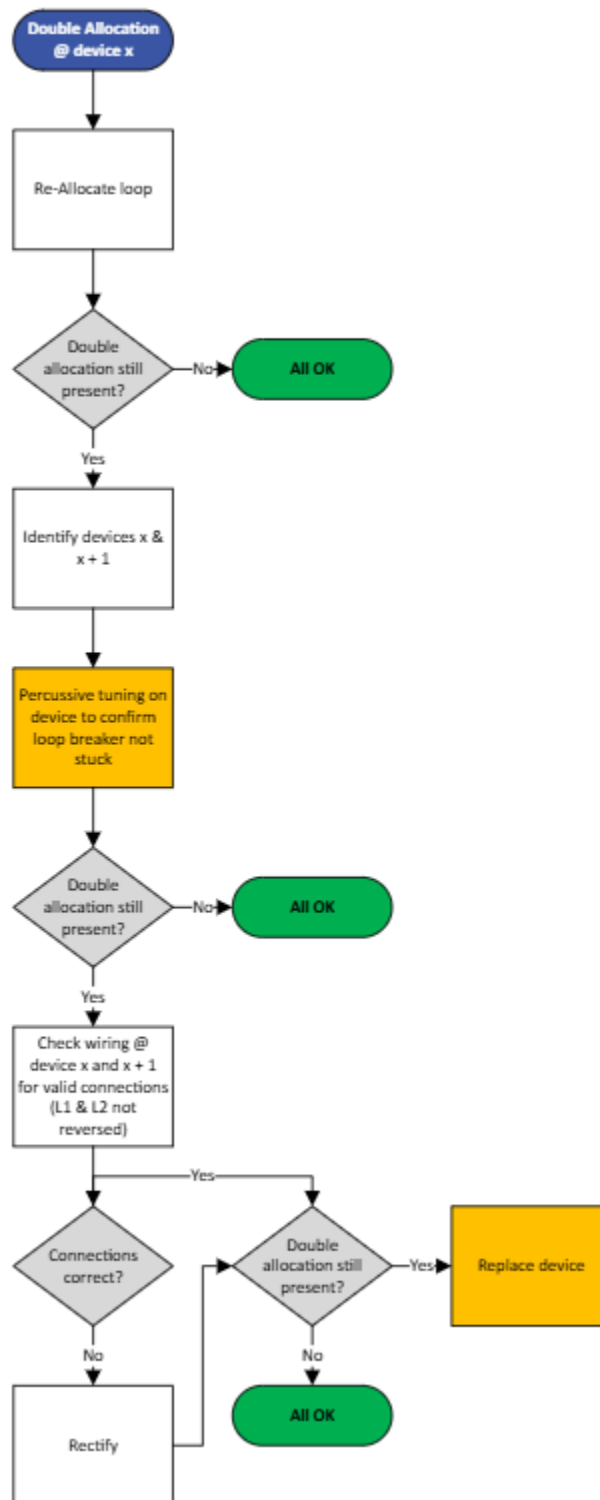
Short Circuit Fault



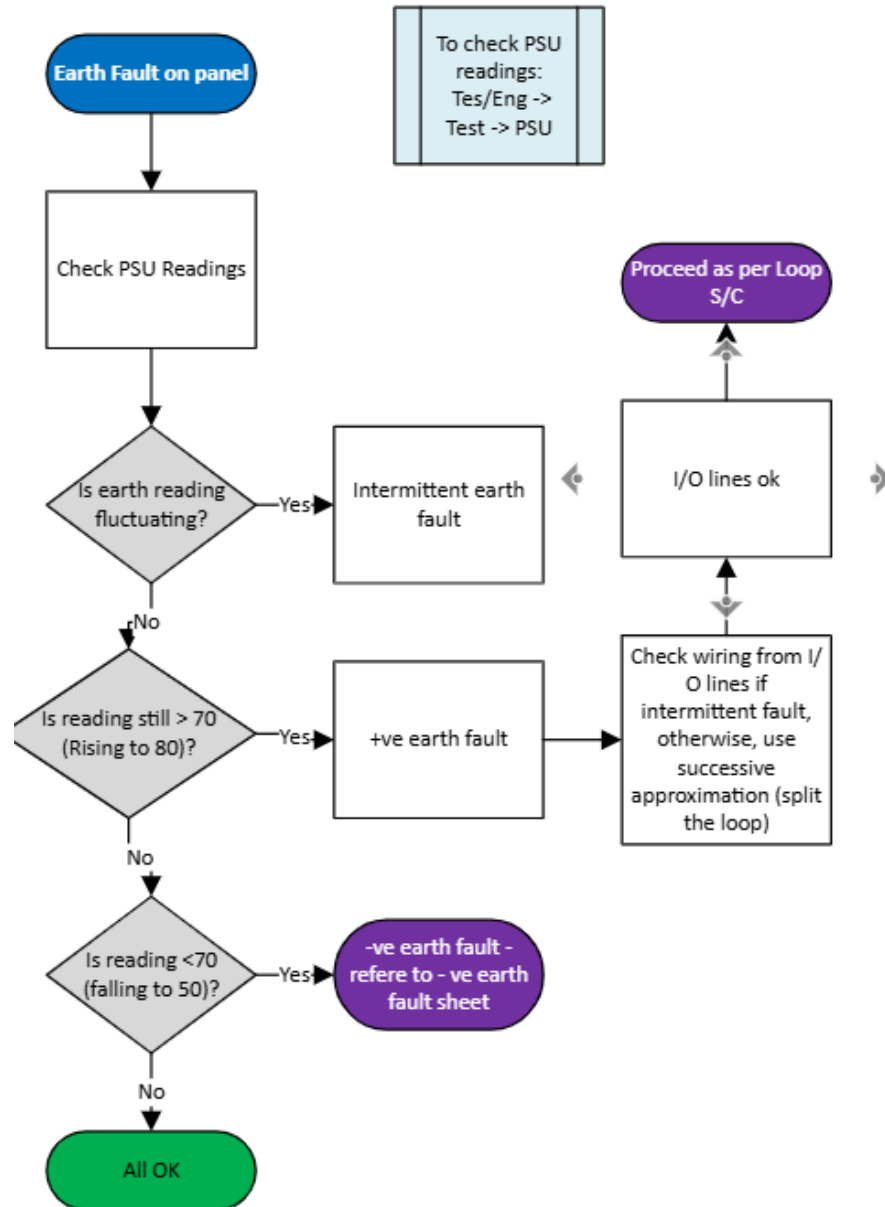
Loop Split (O/C) Fault



Double Allocation



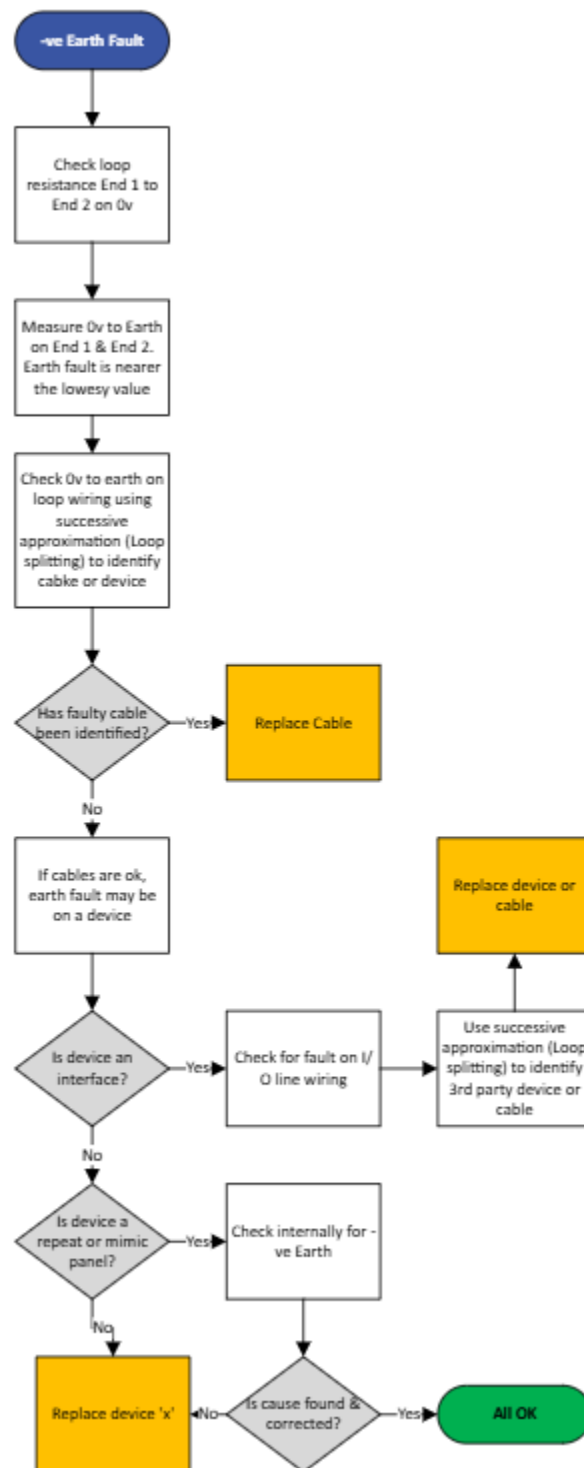
Earth Fault



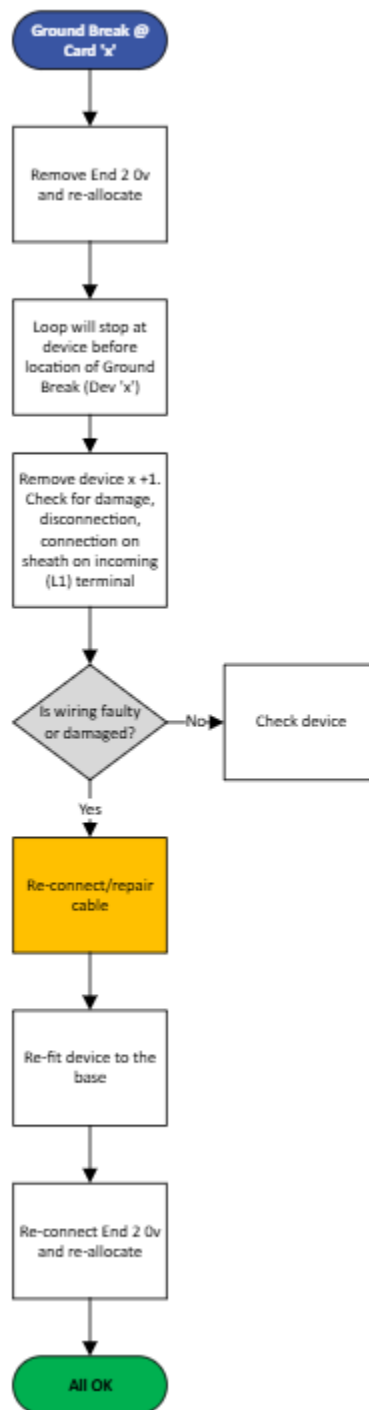
PSU Interpretation

	230v output	DC output after Regulation	Batt 1 Load Test	Batt 1 Off Load	Batt 2 Load Test	Batt 2 Off Load	Master Alarm 1	Master Alarm 2	Earth Monitor	Loop Voltage
	Mns	DC	Bt1	BC1	Bt2	BC2	MA1	MA2	Ear	LpV
4 - 6 Loop panel	214	141	206	206	0	0	127	74	60	221
2 Loop panel	206	138	200	203	0	0	74	74	65	197

-ve Earth Fault



Ground Break



Hardware Fault

